

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (E&E) (Credit System) Degree Examinations
May - June 2023

20EEE164 - AUTOMOTIVE ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss in detail the uses of electronics in development of vehicles.	8	L*2	01	1
b)	Explain the working principle of an engine crankshaft angular position sensor.	6	L1	02	2
c)	Elaborate working of a magnetic reluctance position sensor.	6	L1	02	2
2. a)	With neat a diagram, explain the working of Hall-Effect position sensor.	8	L1	02	2
b)	Elaborate main functions of electronic control unit (ECU).	6	L2	01	1
c)	Explain the process of controlling high and low beam of head lamp.	6	L2	01	1
3. a)	With neat diagram explain working of electronic stability program (ESP) used in vehicle.	8	L2	01	1
b)	Explain the working of automotive engine control actuators.	8	L2	02	2
c)	With neat diagram explain the construction of exhaust gas sensor filament.	4	L1	02	2
Unit – II					
4. a)	With neat diagram explain the different layers used in AUTOSAR.	8	L2	04	2,3
b)	Discuss the following with respect to Flexray protocol. i. Protocol layer ii. Message format	8	L2	03	2,3
c)	Explain data formatting of LIN protocol.	4	L1	03	2,3
5. a)	Explain the following in CAN protocol. i. Protocol layer ii. Voltage level of signal iii. Message format	10	L2	03	2,3
b)	Elaborate AUTOSAR architecture and methodology used in vehicle.	10	L2	04	2,3
6. a)	Explain the design process AUTOSAR used in vehicle.	8	L2	04	2,3
b)	With neat diagram elaborate LIN bus protocol.	8	L1	03	2,3
c)	List the applications of CAN bus.	4	L1	03	2,3
Unit – III					
7. a)	Explain role of microcomputer in electronic system design.	10	L2	05	3
b)	Elaborate various types of interrupts used in vehicle control.	10	L2	05	3
8. a)	Explain various semiconductor memories used in automotive electronics.	10	L1	05	3
b)	Elaborate operating conditions, design and data processing of digital modules present in control unit of a vehicle.	10	L2	05	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
